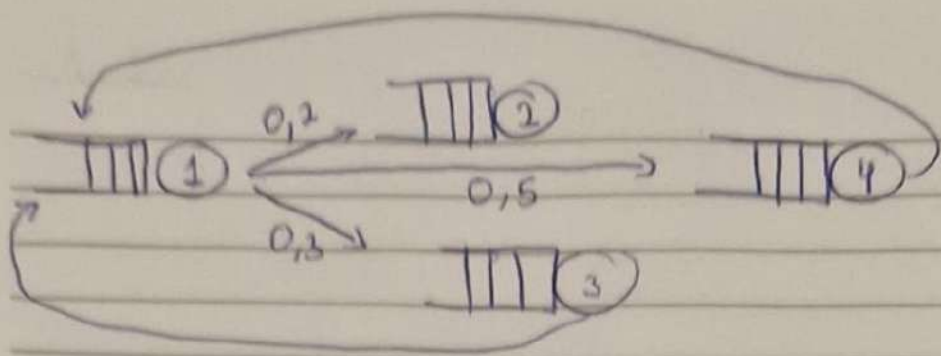




$$S_1 = 0.25$$

$$S_2 = 1.5$$

$$S_3 = 1.25$$

$$S_4 = 0.5$$

$$n_1(29) = 1.8740$$

$$n_2(29) = 3.5212$$

$$n_3 = ?$$

$$n_4(29) = 8.3977$$

$$V_1 = 1$$

$$V_2 = 0.2 V_1 = 0.2$$

$$V_3 = 0.3 V_1 = 0.3$$

$$V_4 = 0.5 V_1 + V_2 = 0.5 + 0.2 = 0.7$$

$$n_1 + n_2 + n_3 + n_4 = 29$$

$$1.8740 + 3.5212 + n_3 + 8.3977 = 29$$

$$29 - n_3 = 13.7929$$

$$n_3 = 29 - 13.7929$$

$$n_{3(29)} = 15.2071$$

$$w_i(30) = S_i (1 + n_i(29))$$

$$w_1(30) = 0.25 (1 + n_1(29)) \rightarrow w_1(30) = 0.25 (1 + 1.8740) \rightarrow w_1(30) = 0.7181$$

$$w_2(30) = 1.5 (1 + 3.5212) \rightarrow w_2(30) = 1.5 (4.5212) = 6.7818$$

$$w_3(30) = 1.25 (1 + 15.2071) \rightarrow w_3(30) = 1.25 (16.2071) = 20.258875$$

$$w_4(30) = 0.5 (1 + 8.3977) \rightarrow w_4(30) = 0.5 (9.3977) = 4.69885$$

$$d_0(N) = \frac{N}{\sum w_i V_i} \quad d_0(30) = 30 / (0.7181 \cdot 1 + 6.7818 \cdot 0.2 + 20.258875 \cdot 0.3 + 4.69885 \cdot 0.7)$$

$$d_0(30) = 30 / (0.7181 + 1.35636 + 6.0776625 + 3.289195)$$

$$d_0(30) = 30 / 11.4413175 = 2.6220756482$$

$$d_i = d_0 V_i$$

$$d_1 = 2.6220756482$$

$$d_2 = 0.5244151296$$

$$d_3 = 0.7866226945$$

$$d_4 = 1.8354529537$$

$$w_i = d_i S_i$$

$$w_1 = 0.65551891205$$

$$w_2 = 0.7866226944$$

$$w_3 = 0.983278368125$$

$$w_4 = 0.91772647685$$

Fila com menor utilização é Q1, com $\pm 65,55\%$ de uso.